



ST BENEDICT'S

SUBJECT	Chemistry
GRADE	11
EXAMINER	Mrs Govender
MODERATOR	Mr Ntombela
DATE	23 July 2025
MARKS	135
DURATION	120 minutes

%

LEARNER NAME: **MEMO**

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 16 pages and a Data Sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment.
4. Remember to write your name and surname in the space provided above AND circle your teachers name.
5. Use the data and formulae sheet whenever necessary.
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
8. Answers must be expressed in decimal format, not left as proper fractions.
9. Where appropriate, express answers to TWO decimal places.
10. It is in your own interest to write legibly and to present your work neatly.
11. Blank pages and blank graphs are included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

Question	1	2	3	4	5	6	7	8	TOTAL
Mark									
Moderated Mark									
Question Total	15	17	18	19	17	14	14	21	135

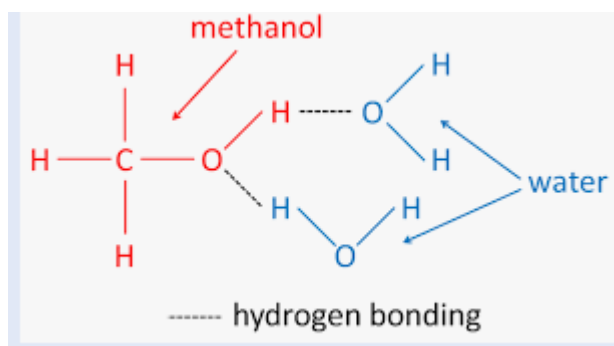
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Learner's Signature:

Teacher's Signature:

QUESTION 1 SHARING IS CARING (SOMETIMES)**15 MARKS**

- 1.1.1 Random movement of electrons✓ result in a temporary dipole being set up in one atom✓ Which causes the electrons in the neighbouring atoms to disperse creating an induced dipole. ✓ The two dipoles are then able to attract each other ✓ (4)
- 1.1.2 Both have London forces between their molecules✓ methane is a larger molecule with more electrons ✓thus it forms a larger temporary dipole ✓ London forces in methane are stronger ✓ more energy is needed to overcome the IMF in methane ✓ hence it has a higher MP (5)
- 1.1.3 Hydrogen sulphide ✓ (1)
- 1.1.4 It is asymmetrical✓ OR -uneven electron cloud distribution and forms a dipole ✓ (2)



Shows two bonds with one methanol ✓✓

Label hydrogen bonds correctly ✓

1.1.5

QUESTION 2 TIME TO GET STEAMY
17 MARKS

2.1.1 $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ (3)

RHS ✓ LHS ✓ BALANCING ✓ **only if LHS= RHS**

2.1.2 A.

MgO forms an ionic crystal lattice ✓ with many ✓ ionic bonds

And strong electrostatic forces of attraction between cations and anions ✓

A lot of energy is required to overcome the strong intermolecular forces ✓ (4)

B. (2)

Ions are held in fixed positions ✓ by strong ionic bonds and are unable to move ✓

2.2

2.2.1 POLAR ✓ COVALENT ✓ BONDS (2)

2.2.2 Water can form 4 ✓ hydrogen bonds whereas HF can only form 2. ✓ (2)

2.3

Iodine is a larger molecule with more electrons ✓

It forms a larger temporary dipole ✓

The forces in iodine are stronger than the forces in methanol ✓

More energy is needed to overcome the stronger forces in iodine. ✓

Hence higher boiling point (4)

QUESTION 3 LIMITING REACTANTS-THE PARTY POOPER**18 MARKS**

3.1.1 (1)

The mass in grams of one mole of that substance. ✓

3.1.2 (5)

Propanol

oxygen

$$M_r = 3(12) + 8(1) + 16$$

$$M_r = 16(2)$$

$$= 60\text{g/mol}$$

$$= 32\text{ g/mol}$$

$$N = \frac{m}{M} = \frac{10}{60} = 0,17\text{ mol} \quad \checkmark$$

$$N = \frac{m}{M} = \frac{50}{32} = 0,15625\text{ mol} \quad \checkmark$$

Ratio propanol: oxygen

$$1:5 \quad \checkmark$$

$$0,17:x$$

$$X = 0,85\text{ mol of O}_2 \quad \checkmark$$

There is only 0,17 mol of propanol available but 0,85 mol is required Therefore propanol is the limiting reactant ✓

3.1.3 (3)

$$\text{Mol of unreacted O}_2 = 1,5625 - 0,85 = 0,7125\text{mol} \quad \checkmark$$

$$\text{Number of O atoms} = 2 \times 0,7125 \times 6.02 \times 10^{23} \quad \checkmark = 8,58 \times 10^{23} \text{ O atoms} \quad \checkmark$$

3.1.4 (3)

Propanol : carbon dioxide

$$1:3 \quad \checkmark$$

$$0,17:X$$

$$X = 0,51\text{ mol} \quad \checkmark$$

$$N = \frac{V}{V_m}$$

$$V = 0,51 \times 22,4 = 11,42\text{ dm}^3 \quad \checkmark$$

3.2

$$n = \frac{v}{V_m} = \frac{4,48}{22,4} = \mathbf{0,2 \text{ mol}}$$

Ratio 2:1

X:0,2

X = 0,4 mol of KF_2

$$m = 0,4 \times (39 + 19)$$

$$= 23,2 \text{ g}$$

$$\% \text{ purity} = \frac{\text{pure mass}}{\text{impure mass}} \times 100$$

$$= \frac{23,2}{25} \times 100$$

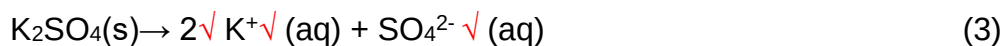
$$= 92,80 \%$$

(6)

QUESTION 4 THE CHEMICAL COCKTAIL HOUR

19 MARKS

4.1.1



4.1.2

$$+ 6\sqrt{\sqrt{\quad}} \quad (2)$$

4.1.3 Calculate the total concentration of K^+ ions when these are mixed. (5) K_2SO_4 KCl

$$n = cv$$

$$n = cv$$

$$= 0,4 \times 0,1$$

$$= 0,3 \times 0,2$$

$$= \mathbf{0,04 \text{ mol}} \quad \sqrt{\quad}$$

$$= \mathbf{0,06 \text{ mol}} \quad n[\text{K}^+] = 0,06 \text{ mol} \quad \sqrt{\quad}$$

$$n[\text{K}^+] = 2\sqrt{\quad} \times 0,04 = 0,08 \text{ mol} \quad C_{\text{new}} = \frac{0,08 + 0,06}{0,3\sqrt{\quad}}$$

$$= \mathbf{0,47 \text{ mol.dm}^{-3}} \quad \sqrt{\quad}$$

4.2

4.2.1 A solution of known concentration $\sqrt{\sqrt{\quad}}$ (2)

$$4.2.2 \quad c = \frac{m}{MV} \sqrt{\quad}$$

$$m = 0,4 \times 0,25\sqrt{\quad} \times (24,3 + 32 + 2)$$

$$= \mathbf{5,83 \text{ g}} \quad \sqrt{\quad} \quad (3)$$

4.2.3 (1)

Volumetric flask $\sqrt{\quad}$

4.2.4 (1)

Funnel $\sqrt{\quad}$

4.2.5 (2)

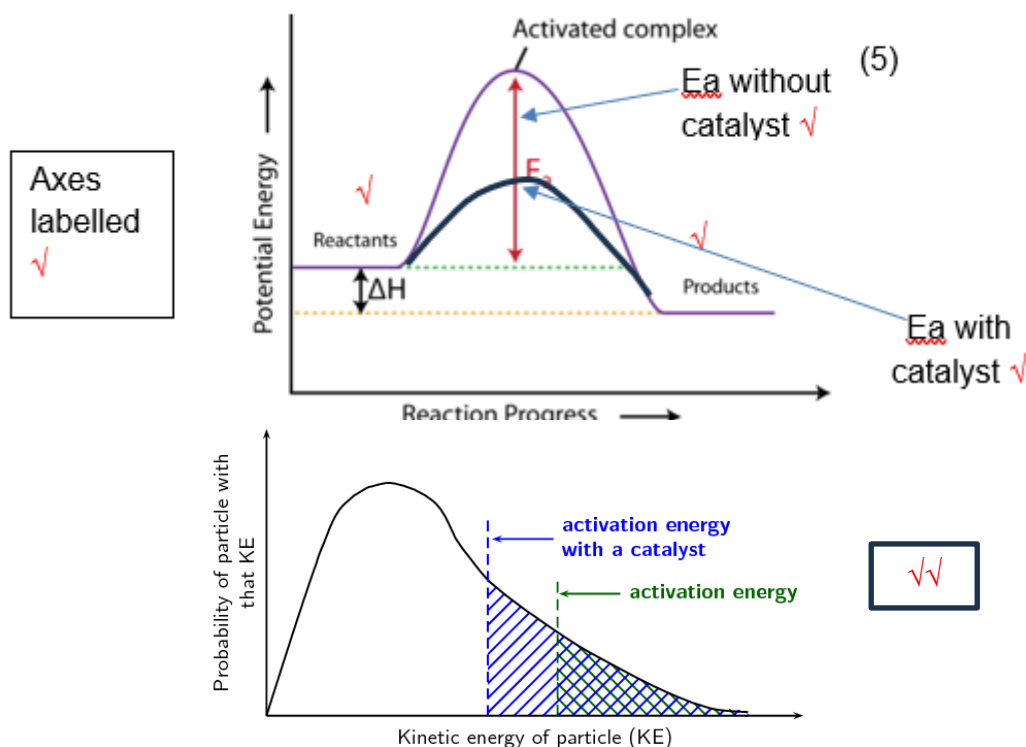
C	F	A	B	D $\sqrt{\sqrt{\quad}}$		
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Minus 1 for every letter in the incorrect order

QUESTION 5 FEELING THE CHILL

14MARKS

5.1



5.2

5.2.1 Number of particles in the reaction. ✓✓ (2)

5.2.2 "Catalysts speed up the reactions by increasing the energy of molecules." "Do you agree. Circle YES or **NO**. ✓ (1)

Support your answer with reference to theory and the graph. (2)

Catalysts lower the activation energy by providing an alternate pathway for the reaction to take place. ✓

More particle have $E_k \geq E_a$. Energy of particles stay the same. ✓

5.2.3

Lower peak ✓

Higher at the end -more effective collisions ✓ (2)

QUESTION 6 17 MARKS

6.1 change in concentration/volume of reactants or products ✓ per unit time ✓
(2)

6.2 $\text{Rate} = \frac{\Delta v}{\Delta t}$ ✓ $= \frac{50}{40} = 1,25 \text{ cm}^3 \cdot \text{s}^{-1}$ ✓ **unit mark is independent** (3)

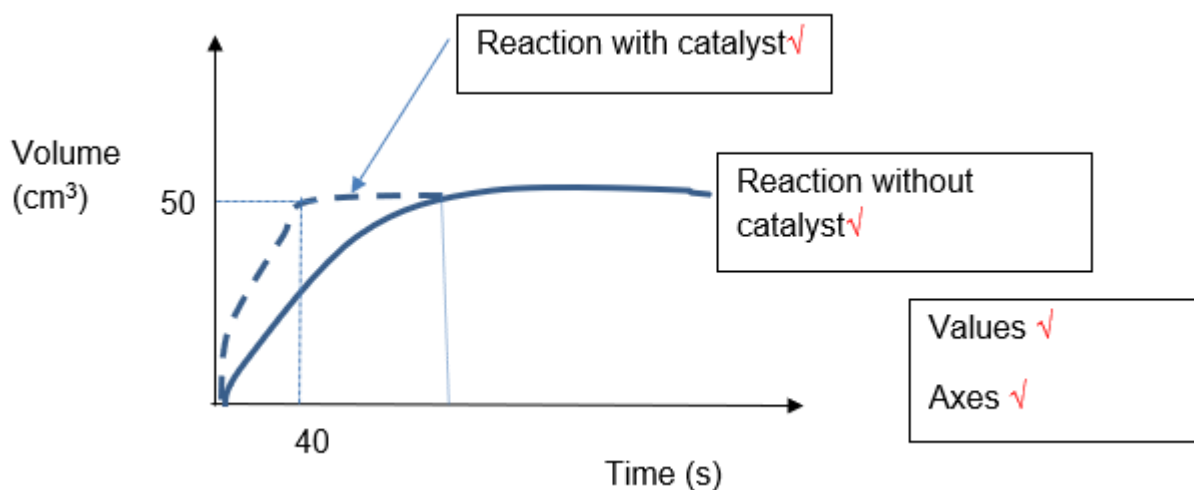
6.3

6.3.1 Unchanged. ✓ Without the catalyst the reaction is slower, but will eventually make 50 cm^3 ✓ (2)

6.3.2 The reactions will end in 40 s with a volume of 50 cm^3 ✓

Rate increases because the time taken decreases ✓ (2)

6.4 Sketch a graph of volume of H_2 versus time for both trials on the same axes. Include all



6.5

Increasing the temperature increases the average kinetic energy of the particles ✓

More particles have kinetic energy $\geq$ activation energy ✓

More collisions occur ,hence more effective collisions occur per unit time ✓

Rate increases ✓ (4)

QUESTION 7 WHEN CHEMISTRY JUST HAPPENS**14 MARKS**

7.1 Write the following reactions: (6)

Oxidation half reaction	$\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^-$ ✓
Reduction half reaction	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ ✓
Net ionic reaction	$2\text{Al} \rightarrow 2\text{Al}^{3+} + 6\text{e}^-$ $6\text{H}^+ + 6\text{e}^- \rightarrow 3\text{H}_2$ $2\text{Al} + 6\text{H}^+ \rightarrow 2\text{Al}^{3+} + 3\text{H}_2$ RHS ✓ LHS ✓ BAL ✓

7.2 Write the formulae of the: (2)

Reducing agent	Al ✓
Spectator ion	Cl^- ✓

7.3 Explain why this reaction is spontaneous. (4)

Al IS A STRONGER REDUCING AGENT ✓ THAN H_2 , ✓

Al is able to reduce H^+ ions ✓ to H_2 ✓

OR

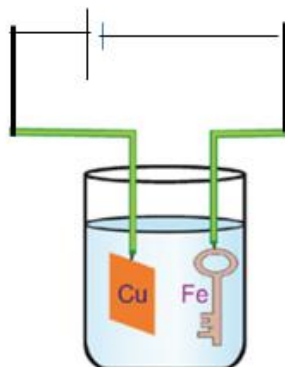
H^+ is a stronger oxidising agent ✓ than Al^{3+} ✓ so H^+ is able to reduce ✓ Al to Al^{3+} ✓

7.4 State what happens to a reducing agent during a reaction, in terms of oxidation number and electron transfer. (2)

Oxidation number	Increases ✓
Electron transfer	Donates/loses electrons ✓

QUESTION 8 WHEN ELECTRICITY FORCES A BREAK UP 21 MARKS

- 8.1 An application of electrolysis is shown below, in which a key is electroplated with copper.



- 8.1.1 What energy conversion takes place in this electrolytic cell? (2)

electrical to chemical ✓✓

- 8.1.2 On the diagram, draw in the battery needed for the cell to work correctly. (2)

- 8.1.3 Write the chemical formula of a substance that needs to be dissolved in water to form the electrolyte for this electroplating. (2)

Cu SO₄ OR Cu(NO₃)₂ ✓✓

- 8.1.4 Write the half reaction that occurs at the key, AND state whether it is the oxidation or reduction half reaction. (3)

$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ - reduction ✓

RHS ✓ LHS ✓

- 8.2 Name the material used to make the electrodes in this cell. (2)

CARBON ✓ or platinum or graphite

- 8.3 State two reasons why this material is used to make the electrodes (2)

Cheap, ✓ easy to source ✓ inert ✓, good conductor ✓ high melting point

- 8.4 (2)

To overcome the strong electrostatic forces ✓ so that ions may be released from their fixed positions and are able to move, thereby conducting electricity ✓

- 8.4.1 _____ electrons ✓ _____ (1)

- 8.4.2 _____ electrons ✓ _____ (1)

8.4.3 ___ions ✓ _____ (1)

8.5 Complete the table showing details of the electrolysis of molten lead (II) bromide. (3)

	Negative electrode	Positive electrode
Observations at the electrode	Silvery grey liquid forms	Bubbles of red brown gas ✓
Half Reaction	$\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$ ✓	$2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{e}^-$ ✓

END OF ASSESSMENT